Software Requirements Specification (SRS)

Boarding House Management & Payment System

Version: 1.0

Prepared For: Software Development Project

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1. Introduction

1.1 Purpose

The Boarding House Management & Payment System is a web-based application designed to help boarding house owners manage members, monthly expenses, payment records, receipt verification, notifications, and communication.

The system provides a centralized platform where owners can manage boarding house operations while members can monitor their payment status, upload payment receipts, and receive important notifications.

The system does not process online payments in Version 1. It only records and verifies payments made externally.

1.2 Scope

The system shall provide:

- Boarding house creation and management
- Member management
- Monthly expense management
- Automatic payment calculation
- Partial payment handling
- Receipt upload and verification
- Outstanding balance tracking
- Carry-forward balance calculation
- Birthday reminders

- Notification management
- Announcement management
- Payment reports and dashboards

Future versions may support online payments, maintenance requests, visitor management, inventory tracking, and mobile applications.

1.3 Intended Users

Owner

Responsible for managing boarding houses, members, expenses, and payment verification.

Member

Responsible for viewing dues, uploading receipts, and tracking payments.

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2. Overall Description

2.1 Product Perspective

The system is a standalone web application consisting of:

- Frontend Web Application
- Backend API Server
- Database Server
- Notification Service
- File Storage for Receipts

2.2 User Roles

Owner

Permissions:

- Create boarding houses
- Manage members

- Manage expenses
- Set due dates
- Verify payments
- View reports
- Publish announcements

Member

Permissions:

- View dues
- Upload receipts
- View payment history
- View notifications
- View profile information

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3. Functional Requirements

FR-01 User Registration and Authentication

Description

The system shall allow users to create accounts and securely log in.

Inputs

- Email
- Password

Outputs

- Successful login
- Error messages

FR-02 Boarding House Management

Description

The owner shall be able to create and manage boarding houses.

Functions

- Create boarding house
- Edit boarding house
- View boarding house details

Data

- House Name
- Created Date
- Owner

FR-03 Member Management

Description

The owner shall manage member records.

Member Information

Personal Information

- Full Name
- Date of Birth
- Gender
- Profile Photo

Contact Information

- Contact Number
- Email Address
- Address

Emergency Information

- Emergency Contact Name
- Emergency Contact Number

- Relationship

Boarding Information

- Joining Date
- Room Number (Optional)
- Status

FR-04 Member Status Management

Description

The owner shall be able to activate or deactivate members.

Rules

Inactive members:

- Shall not be included in future billing calculations.
- Shall retain historical records.

FR-05 Monthly Expense Management

Description

The owner shall enter monthly expenses.

Expense Categories

- Rent
- Electricity
- Water
- Internet
- Other Expenses

Calculation

Monthly Total = Sum of all expenses

Member Due = Monthly Total / Active Member Count

FR-06 Monthly Billing Generation

Description

The system shall generate monthly bills.

Rules

Bills shall be generated for active members only.

Each bill shall contain:

- Month
- Due Amount
- Due Date
- Previous Outstanding Balance
- Total Amount Due

FR-07 Carry-Forward Outstanding Balance

Description

Outstanding balances shall automatically carry forward.

Example

Previous Outstanding = Rs. 1,000

Current Due = Rs. 6,000

Total Due = Rs. 7,000

FR-08 Payment Submission

Description

Members shall submit payment receipts.

Supported Formats

- JPG

- JPEG
- PNG
- PDF

Data

- Receipt File
- Payment Amount
- Remarks (Optional)

FR-09 Partial Payments

Description

Members may pay part of the required amount.

Example

Due = Rs. 6,000

Paid = Rs. 5,000

Outstanding = Rs. 1,000

Status = Partially Paid

FR-10 Multiple Payments

Description

Members shall be allowed to make multiple payments for the same billing period.

Example

Payment 1 = Rs. 2,000

Payment 2 = Rs. 2,000

Payment 3 = Rs. 2,000

Total Paid = Rs. 6,000

Status = Paid

FR-11 Receipt Verification

Description

Owners shall verify payment receipts.

Actions

- Approve
- Reject

Rejection

Owner shall provide a rejection reason.

FR-12 Payment Status Management

The system shall support:

- Pending
- Under Review
- Partially Paid
- Paid
- Overdue
- Rejected

FR-13 Due Date Management

The owner shall define payment deadlines.

Example

Payment Due Date = 5th of each month

FR-14 Overdue Management

If:

Outstanding Amount > 0

and

Current Date > Due Date

Then:

Status = Overdue

FR-15 Notification System

Description

The system shall provide real-time notifications.

Notification Categories

Financial

- Bill Generated
- Due Reminder
- Outstanding Reminder
- Overdue Notice
- Payment Approved
- Payment Rejected

Social

- Birthday Reminder
- Birthday Notification

Administrative

- Announcements
- Expense Updates

System

- Profile Updates
- Account Activities

FR-16 Birthday Management

Description

The system shall notify members about birthdays.

Notifications

One day before birthday.

On birthday.

FR-17 Announcement Management

Description

Owners shall publish announcements.

Example

Water interruption tomorrow.

Internet maintenance scheduled.

FR-18 Dashboard Management

Owner Dashboard

Display:

- Total Members
- Active Members
- Paid Members
- Partially Paid Members
- Overdue Members
- Outstanding Amount
- Monthly Collection

Member Dashboard

Display:

- Current Due
- Paid Amount
- Outstanding Balance
- Due Date
- Payment Status
- Notifications
- Upcoming Birthdays

FR-19 Reports

Monthly Collection Report

- Expected Income
- Collected Amount
- Outstanding Amount

Member Payment Report

- Member Name
- Total Due
- Total Paid
- Outstanding Balance

Payment History Report

- Month
- Payment Amount
- Status

FR-20 Audit Trail

The system shall preserve historical data.

Rules

Records shall not be permanently deleted.

Members shall be marked as Inactive instead of deleted.

Payments shall retain approval history.

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4. Non-Functional Requirements

Performance

- Dashboard loading time less than 3 seconds.
- Support at least 500 active members per boarding house.

Availability

- System availability: 99%.

Security

- Passwords shall be encrypted.
- Role-based access control shall be implemented.
- Users may access only authorized information.

Reliability

- Payment records shall not be lost.
- Receipt files shall remain accessible.

Usability

- Responsive design.
- Mobile-friendly interface.
- Simple navigation.

Maintainability

- Modular architecture.
- RESTful API design.

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5. Future Enhancements (Version 2)

Maintenance Requests

Members can submit:

- Water Issues
- Electrical Issues
- Internet Issues
- Room Repairs

Visitor Management

Track visitor entries and exits.

Inventory Management

Manage:

- Beds
- Chairs
- Tables
- Cupboards
- Keys

Online Payments

- Payment Gateway Integration
- Bank API Integration

Mobile Application

Android and iOS applications.

Messaging System

Direct communication between owners and members.

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6. Assumptions and Constraints

Assumptions

- Members have internet access.
- Owners verify payment receipts manually.
- Payments are performed outside the system.

Constraints

- No payment gateway integration in Version 1.
- Notifications depend on internet connectivity.
- Equal expense sharing among active members.

End of SRS